

Manual for “Measuring US Industry-Level Productivity Between 1947 and 2023”*

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1 Introduction

We combine data from Jorgenson et al. (2017), Eldridge et al. (2020) and Garner et al. (2025) to extend US industry-level production accounts through 2023. Additionally, we apply the methodology in Jorgenson et al. (2007) to produce industry-level measures of labor productivity and total factor productivity (TFP) for the corresponding period. Our dataset covers 63 industries between 1963 and 2023 and 44 industries between 1947 and 2023. To the best of our knowledge, our dataset offers productivity measures with the most extensive temporal coverage and the highest degree of industry disaggregation for the entire US economy currently available.

2 Data

To compute productivity measures between 1947 and 2023, we rely on three underlying datasets:

- [1] **usa_wk_mar_2017.xlsx**: This is produced by Jorgenson et al. (2017) and contains nominal and real values for gross output, capital, labor and intermediate inputs for 65 industries between 1947 and 2014. [DATA LINK]
- [2] **industry-production-account-experimental.xlsx**: This is produced by Eldridge et al. (2020) and contains nominal values and quantity indices of output and intermediate inputs as well as nominal values and quantity indices of different types of capital and labor services. This is in addition to the number of hours worked in each industry. The data cover 63 industries between 1963 and 2016. The dataset also covers the period between 1947 and 1963 for 44 industries. [DATA LINK]

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- [3] **industry-production-account-capital.xlsx**: This dataset produced by Garner et al. (2025) contains nominal values and quantity indices for output as well as different intermediate inputs, capital services, labor services and hours worked for 63 industries between 1997 and 2023. [DATA LINK]

For the version of our dataset covering 63 industries between 1963 and 2023, we use nominal gross output and factor compensation from Garner et al. (2025) from 1997 onward. Before 1997, we link the nominal capital, labor and intermediate input series from Jorgenson et al. (2017) to those of Garner et al. (2025) using industry- and input-specific ratios in 1997. For each industry j and input $X \in \{K, L, I\}$, the linked pre-1997 nominal series is

$$X_{j,t}^{\text{linked}} = X_{j,t}^{\text{KLEMS}} \left(\frac{X_{j,1997}^{\text{BEA-BLS}}}{X_{j,1997}^{\text{KLEMS}}} \right), \quad t < 1997.$$

Gross output is then constructed as the sum of the linked capital, labor and intermediate input series, while value added is the sum of capital and labor compensation. We use the quantity indices from Eldridge et al. (2020) for the period between 1963 and 1997 and supplement these with the quantity indices from Garner et al. (2025) for the period between 1998 and 2023. Data on total hours worked are obtained from Eldridge et al. (2020) for the period between 1963 and 1997 and from Garner et al. (2025) for the period between 1998 and 2023.

Although Jorgenson et al. (2017) provide nominal gross output and factor compensation for the industries underlying the 63-industry classification from 1947 onward, the historical quantity indices needed to produce productivity measures between 1947 and 1963 cover only 44 industries Eldridge et al. (2020). We therefore provide productivity measures for 44 industries between 1947 and 2023.

The list of 44 industries groups some of the 63 industries together rather than creating an entirely new industry classification system. We construct the 44-industry nominal series by aggregating the linked 63-industry KLEMS series before 1997 and the BEA-BLS series from Garner et al. (2025) from 1997 onward. To create the accompanying input quantity and productivity indices, we use the underlying data from Eldridge et al. (2020) for the 44 industries between 1947 and 1963. Since the 44 industries group some of the 63 industries together, we extend the quantity growth rates for the 44 industries beyond 1963 by aggregating the corresponding 63-industry growth rates using Törnqvist-weighted nominal value-added shares.

3 Methodology

The output of each industry (Q) is produced by combining three factors of production. These are capital services (K), labor services (L) and intermediate inputs (I). In this section, we discuss how we compute quantity indices for these three types of inputs for each industry, and how we use these to create output- and value-added productivity measures. We follow a similar productivity measurement strategy to Jorgenson et al. (2007).

3.1 Input Quantity Growth

Not all labor services are the same. Labor services provided by college-educated individuals will differ from the labor services provided by non-college-educated individuals. Likewise, capital services and intermediate inputs are also aggregations of different types of capital services (such as IT equipment capital and software capital) and intermediate inputs (such as energy and materials), respectively. Hence, it is necessary to introduce notation to represent the different types of factors of production. Let subscript k denote the type of capital such that $k \in \{1, \dots, \mathcal{K}\}$. Similarly, let the types of labor services and intermediate inputs be denoted by $l \in \{1, \dots, \mathcal{L}\}$ and $i \in \{1, \dots, \mathcal{I}\}$, respectively. Finally, let subscript t denote the year.

To compute the growth rate of capital, labor and intermediate input quantities used to produce output in a given industry, it is necessary to aggregate the quantity growth rates of the different types of capital services, labor services and intermediate inputs, respectively. This implies

$$\begin{aligned}\Delta \ln K_t &= \sum_k \bar{\omega}_{k,t} \Delta \ln K_{k,t} \\ \Delta \ln L_t &= \sum_l \bar{\omega}_{l,t} \Delta \ln L_{l,t} \\ \Delta \ln I_t &= \sum_i \bar{\omega}_{i,t} \Delta \ln I_{i,t}\end{aligned}$$

Letting the price of each type of capital, labor and intermediate input be denoted by P_k , P_l and P_i , respectively, the growth rate of each type of capital, labor and intermediate input is weighted by the two-period average nominal share (hereafter referred to as the Törnqvist weight) of total factor compensation paid to each type of factor:

$$\begin{aligned}\bar{\omega}_{k,t} &= 0.5 \times (\omega_{k,t} + \omega_{k,t-1}) \quad \text{where} \quad \omega_{k,t} = \frac{P_{k,t} K_{k,t}}{\sum_k P_{k,t} K_{k,t}} \\ \bar{\omega}_{l,t} &= 0.5 \times (\omega_{l,t} + \omega_{l,t-1}) \quad \text{where} \quad \omega_{l,t} = \frac{P_{l,t} L_{l,t}}{\sum_l P_{l,t} L_{l,t}} \\ \bar{\omega}_{i,t} &= 0.5 \times (\omega_{i,t} + \omega_{i,t-1}) \quad \text{where} \quad \omega_{i,t} = \frac{P_{i,t} I_{i,t}}{\sum_i P_{i,t} I_{i,t}}\end{aligned}$$

These growth rates are then chained to produce the input quantity indices.

3.2 Output-Based Productivity Growth

We produce two output-based measures of productivity: output per hour worked and output TFP. This subsection describes how we compute the growth rates of each of these productivity measures, which we later chain to produce the indices.

Starting with the growth rate of output quantity per hour worked, this is simply

$$\Delta \ln \left(\frac{Q_t}{H_t} \right) = \Delta \ln Q_t - \Delta \ln H_t$$

where H_t is the total number of hours worked by college- and non-college-educated workers to produce output Q_t . Meanwhile, the growth rate of output TFP is

$$\Delta \ln TFP_t^Q = \Delta \ln Q_t - \bar{\omega}_{K,t} \Delta \ln K_t - \bar{\omega}_{L,t} \Delta \ln L_t - \bar{\omega}_{I,t} \Delta \ln I_t$$

using Törnqvist-weighted factor compensation shares of total expenditure:

$$\bar{\omega}_{K,t} = 0.5 \times (\omega_{K,t} + \omega_{K,t-1}) \quad \text{where} \quad \omega_{K,t} = \frac{P_{K,t}K_t}{P_{K,t}K_t + P_{L,t}L_t + P_{I,t}I_t}$$

$$\bar{\omega}_{L,t} = 0.5 \times (\omega_{L,t} + \omega_{L,t-1}) \quad \text{where} \quad \omega_{L,t} = \frac{P_{L,t}L_t}{P_{K,t}K_t + P_{L,t}L_t + P_{I,t}I_t}$$

$$\bar{\omega}_{I,t} = 0.5 \times (\omega_{I,t} + \omega_{I,t-1}) \quad \text{where} \quad \omega_{I,t} = \frac{P_{I,t}I_t}{P_{K,t}K_t + P_{L,t}L_t + P_{I,t}I_t}$$

3.3 Value-Added-Based Productivity Growth

We produce two value-added-based measures of productivity: value-added quantity per hour worked and value-added TFP. To compute these, we must start by computing a measure of value-added quantity growth. This is

$$\Delta \ln VAQ_t = \frac{\Delta \ln GO_t - \bar{\omega}_{I,t} \Delta \ln I_t}{\bar{\omega}_{VA,t}}$$

using the Törnqvist-weighted nominal value-added share of output:

$$\bar{\omega}_{VA,t} = 0.5 \times (\omega_{VA,t} + \omega_{VA,t-1}) \quad \text{where} \quad \omega_{VA,t} = 1 - \omega_{I,t}$$

With this, we can compute the growth rate of value-added quantity per hour worked as

$$\Delta \ln \left(\frac{VAQ_t}{H_t} \right) = \Delta \ln VAQ_t - \Delta \ln H_t$$

and the growth rate of value-added TFP as

$$\Delta \ln TFP_t^{VA} = \Delta \ln VAQ_t - \bar{\omega}_{K,t}^{VA} \Delta \ln K_t - \bar{\omega}_{L,t}^{VA} \Delta \ln L_t$$

using Törnqvist-weighted nominal capital and labor shares of value added:

$$\bar{\omega}_{K,t}^{VA} = 0.5 \times (\omega_{K,t}^{VA} + \omega_{K,t-1}^{VA}) \quad \text{where} \quad \omega_{K,t}^{VA} = \frac{P_{K,t}K_t}{P_{K,t}K_t + P_{L,t}L_t}$$

$$\bar{\omega}_{L,t}^{VA} = 0.5 \times (\omega_{L,t}^{VA} + \omega_{L,t-1}^{VA}) \quad \text{where} \quad \omega_{L,t}^{VA} = \frac{P_{L,t}L_t}{P_{K,t}K_t + P_{L,t}L_t}$$

References

- Eldridge, L. P., Garner, C., Howells, T. F., Moyer, B. C., Russell, M., Samuels, J. D., Strassner, E. H. & Wasshausen, D. B. (2020), Toward a bea-bls integrated industry-level production account for 1947–2016, *in* ‘Measuring economic growth and productivity’, Elsevier, pp. 221–249.
- Garner, C., Harper, J., Russell, M. & Samuels, J. (2025), Integrated bea-bls industry-level production account, 1997–2023: The sources of u.s. economic growth in the aftermath of the covid-19 recession, Technical report, Bureau of Economic Analysis and Bureau of Labor Statistics.
- Jorgenson, D. W., Ho, M. S. & Samuels, J. D. (2017), Educational attainment and the revival of us economic growth, *in* ‘Education, Skills, and Technical Change: Implications for Future US GDP Growth’, University of Chicago Press, pp. 23–60.
- Jorgenson, D. W., Ho, M. S., Samuels, J. D. & Stiroh, K. J. (2007), ‘Industry origins of the american productivity resurgence’, *Economic systems research* **19**(3), 229–252.